

AI statement

I used AI tools to assist with specific tasks during development:

- Testing Setup: assisted with configuring Vitest and React Testing Library, and provided examples of testing patterns that I adapted for my components.
- Chart Configuration: helped with initial Recharts setup and configuration examples, which I customized for my analytics page requirements.
- Tailwind CSS: Tailwind class combinations for specific layouts and responsive design patterns, which I then adapted to match my design.
- Component Structure: React component patterns and best practices for organizing my code, particularly for setting up the initial project structure with Vite and React Router v7.
- Swagger Documentation: helped generate initial Swagger schema definitions and examples, which I then customized with my actual API endpoints and data models.
- Documentation: helped mostly with the README files and making sure my comments were correct.
- Debugging: When encountering errors (e.g., PostgreSQL connection issues, deployment configuration), I used AI to help interpret error messages and suggest potential solutions.
- Testing: AI helped write the initial test structure for Jest and Supertest, which I then expanded with my specific test cases and validation scenarios.

However, I fully understand all the code in my submission. I can explain:

- Why I chose Context API over Redux for state management (simpler, no extra dependencies, sufficient for auth state)
- How React Router v7's protected routes work with my authentication flow
- The purpose of Axios interceptors for automatic token attachment
- How I implemented form validation with date range checks
- My testing strategy and why I chose Vitest over Jest
- Why I chose JWT over sessions for authentication (stateless, scalable, works well with frontend)
- How the many-to-many relationship works with the trip_participants junction table
- The purpose of each middleware in my Express pipeline (authentication, validation, error handling)
- How parameterized queries prevent SQL injection attacks
- My refactoring decisions and the atomic commits showing the process

I used AI as a learning tool and productivity booster, not as a replacement for understanding. All architectural decisions, database design choices, and all design decisions, component architecture, and problem-solving approaches were my own work.